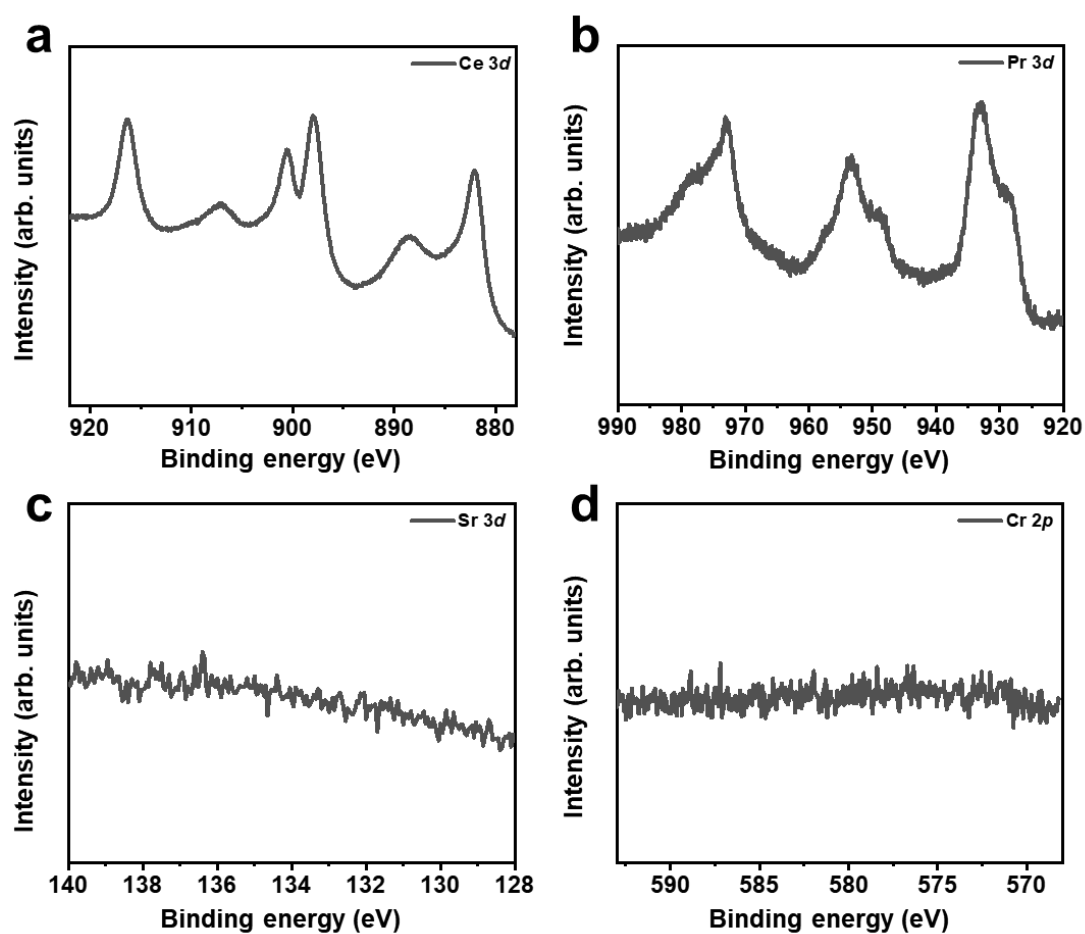
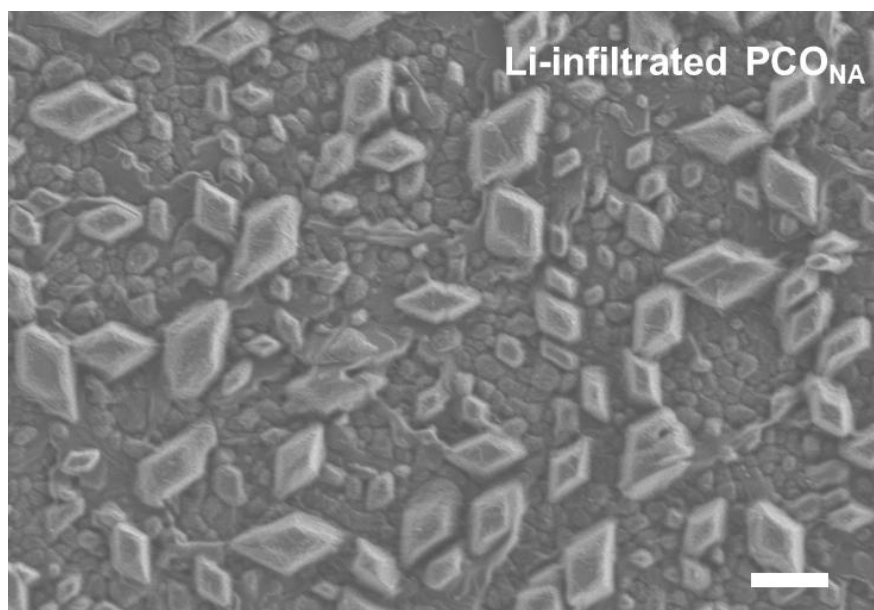


Supplementary Table 2. Absolute conductivity values for the binary oxides (Li₂O and SiO₂) at 600 °C used in this work.

	Conductivity (S/cm) at 600 °C
Li ₂ O (for basic) ¹⁸	- 10 ⁻⁵
SiO ₂ (For acidic) ¹⁹	- 10 ⁻¹²



Supplementary Fig. 1 X-ray photoelectron spectroscopy (XPS) spectra of pristine PCO_{NA} with the different binding energy regions of Ce 3d, Pr 3d, Sr 3d, and Cr 2p spectra, respectively. XPS data exhibits no presence of Sr or Cr elements at the surface which could cause surface degradation in PCO_{NA} sample.



Supplementary Fig. 2 SEM image of Li-infiltrated PCO_{NA} fabricated on Al₂O₃. The PCO_{NA} is observed to be fully covered by Li₂O given infiltration with a concentrated solution. (scale bar, 2 μ m).